

MCS-1: Probability and Statistics (MA6.302)

Monsoon 2026

Lecture 10

September 8th, 2026



Conditional PDFs

- Let X be a continuous random variable with pdf function $f_X(x)$ and $A \in \mathbb{R}$ be such that $P(X \in A) > 0$
- Conditional PDF** of X given $X \in A$ is defined as follows:

$$f_{X|\{X \in A\}}(x) = \begin{cases} \frac{f_X(x)}{P(X \in A)}, & \text{if } X \in A \\ 0, & \text{otherwise} \end{cases}$$

- In this case, we can write the conditional probability as

$$\begin{aligned} P(X \in B | X \in A) &= \int_{x \in B} f_{X|\{X \in A\}} dx = \int_{x \in B} \frac{f_X(x) \mathbb{I}_{[X \in A]}}{P(X \in A)} dx \\ &= \int_{A \cap B} \frac{f_X(x)}{P(X \in A)} dx = \frac{P(X \in A \cap B)}{P(X \in A)} \end{aligned}$$

- $f_{X|\{X \in A\}}(x) \geq 0$ as $f_X(x) \geq 0$, $\forall x$ and $P(X \in A) > 0$.
- $\int_{-\infty}^{\infty} f_{X|A}(x) dx = \int_A \frac{f_X(x)}{P(X \in A)} dx = \frac{P(X \in A)}{P(X \in A)} = 1$
- Conditional PDF is a valid PDF:** non-negative and integrates to one.

Conditional PDFs: Example

- The height X of a randomly picked American woman can be modeled by $X \sim \mathcal{N}(63.7, 2.72)$

Conditional PDFs: Example

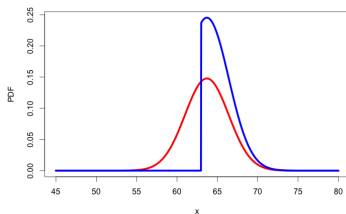
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- The PDF of heights (X) is shown in red.

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- The height X of a randomly picked American woman can be modeled by $X \sim \mathcal{N}(63.7, 2.72)$
- What's the conditional PDF given that the randomly picked woman is at least 63 inches tall?
- The PDF of heights (X) is shown in red.
- The conditional PDF given $X > 63$, shown in blue, is the same shape for $X > 63$... but scaled up to integrate to one.



Conditioning on a different random variable

- So far, we conditioned X on an arbitrary event A , or on a range of values of X .

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- We know we could condition on a range of outcomes of Y , by replacing the arbitrary event A with the event $\{Y \in A\}$

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 - Pick some number, say 0.6777, now generate 100 $\mathcal{N}(0; 1)$ random variables. I will bet a 100\$ that you won't see that number.

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 - Pick some number, say 0.6777, now generate 100 $\mathcal{N}(0; 1)$ random variables. I will bet a 100\$ that you won't see that number.
 - But when you simulate from the standard normal, you will get a 100 different values, right?

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- $f_{X|Y}(x) = \frac{f_{XY}(x,y)}{f_Y(y)}$ given $f_Y(y) > 0$
- What does this mean?

$$\begin{aligned} f_{X|Y}(x)dx &= \frac{f_{XY}(x,y)dxdy}{f_Y(y)dy} \\ &= \frac{P(x \leq X \leq x+dx, y \leq Y \leq y+dy)}{P(y \leq Y \leq y+dy)} \\ &= P(x \leq X \leq x+dx | y \leq Y \leq y+dy) \end{aligned}$$

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- This is a PDF version of our multiplication rule.
- We can extend it to more than 2 random variables:
$$f_{X,Y,Z}(x,y,z) = f_{Z|X,Y}(z|x,y)f_{Y|X}(y|x)f_X(x)$$

Example: Calculating the Conditional PDF

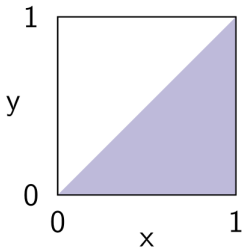
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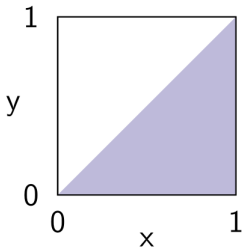
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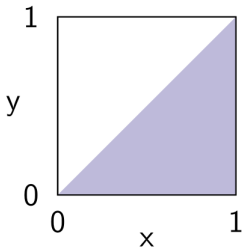
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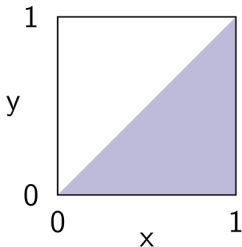
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- What is c ?
- The total area where $x \in [0, 1]$, $y \in [0, x]$ is 0.5, so $c = 2$.

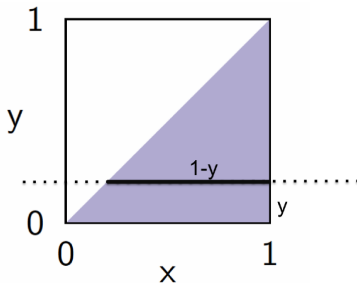
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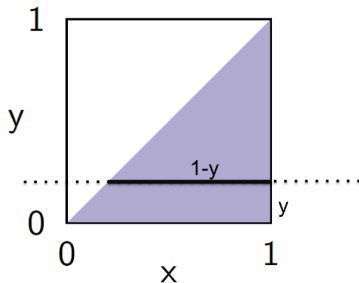
- What is c ?
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- What is the marginal PDF of Y , $f_Y(y)$?

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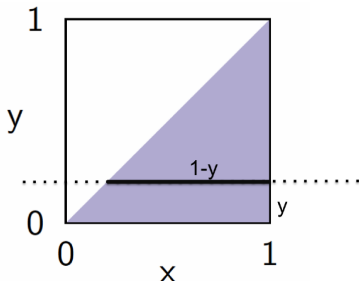
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- $f_Y(y) = \int_0^1 f_{X,Y}(x,y) dx = \int_0^1 2\mathbb{I}[0 \leq x \leq 1; 0 \leq y \leq x] dx = 2 \int_{x=y}^1 dx = 2(1-y)$

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- So, the conditional PDF of X given $Y = y$ is

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \begin{cases} \frac{1}{2(1-y)}, & \text{if } x \in [y, 1] \\ 0, & \text{otherwise} \end{cases}$$

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$$P(X = x) = \sum_{i=1}^n P(X = x|B_i)P(B_i)$$
- How about continuous r.v.'s? Replace $P(X = x|B_i)$ by conditional pdf. $f_X(x) = \sum_{i=1}^n f_{X|B_i}(x)P(B_i)$

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- Pretend Y is a discrete r.v. Then,

$$P(A|Y = y) = \frac{P(Y = y|A)P(A)}{P(Y = y|A)P(A) + P(Y = y|A^c)P(A^c)}$$

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- All that changes for a continuous r.v. is:

$$P(A|Y = y) = \frac{f_{Y|A}(y)P(A)}{f_{Y|A}(y)P(A) + f_{Y|A^c}(y)P(A^c)}$$

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- Without flu, Y is a normal random variable with $\mu = 98.6$ degrees and $\sigma = 0.5$.

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- Body temperature is Y .
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- With flu, Y is a normal random variable with $\mu = 102$ and $\sigma = 2$.

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- With flu, Y is a normal random variable with $\mu = 102$ and $\sigma = 2$.
- My temperature is 100. If A is the event “has flu” and Y is temp.

$$f_{Y|A}(y) = \frac{1}{2\sqrt{2\pi}} e^{-\frac{(y-102)^2}{2 \times 4}}$$

$$f_{Y|A^c}(y) = \frac{1}{0.5\sqrt{2\pi}} e^{-\frac{(y-98.6)^2}{2 \times 0.25}}$$

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- What will be $P(A|Y = y)$?

$$P(A|Y = y) = \frac{f_{Y|A}(y)P(A)}{f_{Y|A}(y)P(A) + f_{Y|A^c}(y)P(A^c)}$$

$$P(A|Y = 100) = \frac{0.2 \times \frac{1}{2\sqrt{2\pi}} e^{-\frac{(y-102)^2}{2 \times 4}}}{0.2 \times \frac{1}{2\sqrt{2\pi}} e^{-\frac{(y-102)^2}{2 \times 4}} + 0.8 \times \frac{1}{0.5\sqrt{2\pi}} e^{-\frac{(y-98.6)^2}{2 \times 0.25}}}$$